

CS231 (Section A)

Surprise Quiz 5 (Bonus Marks: $6 \times 0.05\% = 0.3\%$)

1. `fork()` is called once but returns twice. What value does it return in the child process, and what does it return in the parent process?

- A. Child gets 0; parent gets the child's PID
- B. Child gets the parent's PID; parent gets 0
- C. Both processes receive 0
- D. Both processes receive the same non-zero PID

2. Under Copy-on-Write (COW), what happens immediately after `fork()` returns?

- A. Child gets a full private copy immediately
- B. Both share read-only pages; a writer gets its own copy
- C. Only the parent retains memory access
- D. Child shares parent's memory permanently

3. Besides terminating the process, what else does the `exit()` syscall do?

- A. Suspends the process until the parent responds
- B. Calls `fork()` to notify the shell
- C. Cleans up to reclaim memory, flushes buffers, closes fds
- D. Restarts the process from its entry point

4. What is the key difference between `wait()` and `waitpid()`?

- A. `wait()` is non-blocking, `waitpid()` always blocks
- B. `wait()` only works once, `waitpid()` repeatedly
- C. `wait()` returns void, `waitpid()` returns exit status
- D. `wait()` waits for any child, `waitpid()` targets a specific one

5. What value of counter will be printed by the child and by the parent respectively in Figure-1?

- A. Child prints 1, parent prints 2
- B. Both print 3
- C. Both print 0
- D. Child prints 2, parent prints 1

6. Which statement best describes how the shell works?

- A. Runs in kernel mode (Level 0)
- B. Loops in user mode, spawning a process per command via syscall
- C. Exits after its first command
- D. Needs a reboot for each new command

```
int counter = 0;
int main() {
    int pid = fork();
    if (pid == 0) {
        counter++;
    } else {
        counter += 2;
    }
    printf("Counter = %d\n", counter);
    return 0;
}
```

Figure-1

Answer Key

1. A. Child gets 0; parent gets the child's PID
2. B. Both share read-only pages; a writer gets its own copy
3. C. Cleans up to reclaims memory, flushes buffers, closes fds
4. D. wait() waits for any child, waitpid() targets a specific one
5. A. Child prints 1, parent prints 2
6. B. Loops in user mode, spawning a process per command via syscall